Quantifying the Effects of Failed Coastline Management: A Case Study on Shoreline Erosion in Buxton, North Carolina

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Problem Statement

The Outer Banks is a string of barrier islands and peninsulas that separate the Atlantic Ocean from the mainland of North Carolina. Averaging only a mile in width, this narrow island strand is home to generations of waterman who support North Carolinas commercial and recreational fishing industry and bolster a significant portion of the state's tourism. Ever since the creation of semi-permanent structures along North Carolina's dynamic beach strand known as the Outer Banks, erosion from major storms have been among the top concerns of the residents and homeowners in the area. The center of erosion and damage to human structures has undoubtably been Cape Hatteras, a major storm hotspot, harboring many hurricane and Nor'easter landfalls throughout the years of recorded weather history *(NPS, 2025)*. These storms have brought major impacts to the human infrastructure along the barrier islands, with their impacts centered in Buxton, the town which runs along the Cape Hatteras peninsula. These impacts include damage to Highway 12, the road that runs from north to south, down the northern Outer Banks through Cape Hatteras National Seashore and ending in Ocracoke *(NPS, 2025)*. Additionally, Houses fall into the ocean during the midst of storm surge and king tides, destroying personal property and leeching dangerous sewage and debris into the surf. Just recently, 9 houses fell from tropical storm Imelda impacts in just a 2-day stretch within the Buxton town limits *(NPS, 2025)*. The storm did not even make landfall. In between October 15th and November 15th of 2025, 15 houses fell within a half mile of each other *(NPS, 2025)*. Also, old Naval infrastructure from the World War era has been exposed and is now leaking toxic petroleum products directly into one of the most famous beaches in North Carolina (Cape Hatteras Lighthouse Beach) along the Buxton, NC shoreline *(NPS, 2025)*. Although this year's storms have been typical in severity, strength, and frequency, the past month has shown a notable

rise in major erosion. This project will analyze long-term erosion trends with volumetric and two-dimensional shoreline tools, determine possible causes for the recent uptick, and recommend ways for local government and the NPS to protect the beach strand and nearby structures. Additionally, many past shoreline analysis studies within the area of interest do not use volumetric data to find erosion hotspots. Due to the use of volumetric analysis in this study, factors outside of shoreline recession such as dune erosion and outer sandbar depletion will be accounted for. Thus, providing new insight to what areas may experience severe future erosion due to the depletion of natural elevation and wave energy dispersion.

Background Information

Town and Groin structure in Buxton

Buxton, North Carolina is a small town situated on the peninsula of the Cape Hatteras National Seashore. The eastward facing portion of the peninsula is home to a stretch of motels and residential areas, initially most of these structures were built a distance away from the beach that was thought to be safe in the 1980's. However, erosion in recent years due to a multitude of factors, which will be examined in this study, has introduced beach-neighboring structures to the harsh conditions of the surf-line. South of this residential area lies a National Park Service (NPS) protected area known as lighthouse Beach. ***(Figure 1)***

Figure 1: The study area zoomed in from the east coast of the United States to the Buxton, NC Shoreline.

The Lighthouse beach resides within the National Park Service boundaries for Cape Hatteras National Seashore – meaning there can be no construction within the area that is not a part of the NPS program. The natural landscape of the area consisted of a once tall dune line followed by dense maritime forest vegetation. Extruding from the dune line were three groins which were built to protect the lighthouse and a nearby naval base from erosion. Artificially supported beach strands have long held together beach town infrastructures, from Miami beach to the New Jersey coastline, Groins and Jetties are used to stabilize inlets, limit major erosion events, and protect shore-neighboring buildings. However, these artificially man-made structures require relatively consistent upkeep and structural reinforcement. After the state of North Carolina passed a ban on hard structures for beach strands in 1985 – specifically including a ban on jetties, seawalls, and groins – they neglected to care for the deteriorating groins that reside at Cape Hatteras Lighthouse beach *(NCpedia, 2025)*. In turn, shoreface erosion caused the NPS to move Cape Hatteras Lighthouse hundreds of yards inland from its original position *(NPS, 2025)*. The lighthouse had to be moved in part because no work was able to be done to the failing groins due to the ban on hard structures.

General Erosional Trends

Erosion along the Outer Banks is driven by a disproportionate southward sand migration during the fall, winter, and spring months as a result of Nor'easter, tropical, and subtropical cyclone impacts and landfall *(Ashton et, al.)*. In the summer months, accretional patterns can take place with a predominant northward and inland sand migration. However, the magnitude of this pattern is lesser than that of the southward sediment migration trend seen in all other times of the year *(Ashton et, al.)*. Erosion upstream, or to the north of the groins on Lighthouse Beach, was

limited by the groins ability to hold and stabilize sediment, creating a semi-stable environment for structures to be built.

Methods

Data Collection and Initial Visualization

ArcGIS pro was used in all mapping and data extraction methods. A set of 10 Digital Elevation Models (DEM's), consisting of data from 2009 to 2019, were collected from the *Digital Coast: NOAA Data Viewer (NOAA, 2025)*. The DEM's encompass the shoreline and neighboring structures of Buxton, NC. DEM data collection surveys occurred annually. Consequently, analysis was set for annual volumetric change (study to study). After all 10 DEM's were compiled and symbolized, the "raster the calculator" tool was used to generate raster images that depict difference in elevation between two DEM's.

Figure 2: the area that DEM's were pulled from during the data collection phase.

This process was repeated for every annual analysis window. Using the symbolized raster differences, elevation loss hotspots became visually identifiable. Additionally, the location of fallen houses within the Buxton town limits were inserted as a point layer into the ArcGIS project. Visual analysis of the raster differences along with the base DEM's and the point layer of fallen houses led to the creation of two Areas of Interest (AOI). The Shoreline Zone – encircling the clustered area where 15 houses fell between October 15th, 2025 and November 15th, 2025 - was designed to capture the general trends of the area that has experienced the most residential structure damage (Figure 3). The lighthouse Jetty Beach zone (LHJ Zone) was created

to capture Lighthouse Beach, which is where the deteriorating groins and Cape Hatteras Lighthouse reside (Figure 3).

Figure 3: Shoreline AOI (top) with fallen houses point layer and LHJ AOI (bottom), respectively.

Data Generation and Extraction

To generate data that is exportable into data analysis software, the "extract by mask" geoprocessing tool was used to extract the DEM's from within the AOI's. Following extraction, the "CutFill" Geoprocessing tool was used on the extracted DEM's (figure 4). CutFill organizes

areas of net volumetric loss and gain in a way that allows for the transformation of data into a table that is extractable and formatted for external analysis.

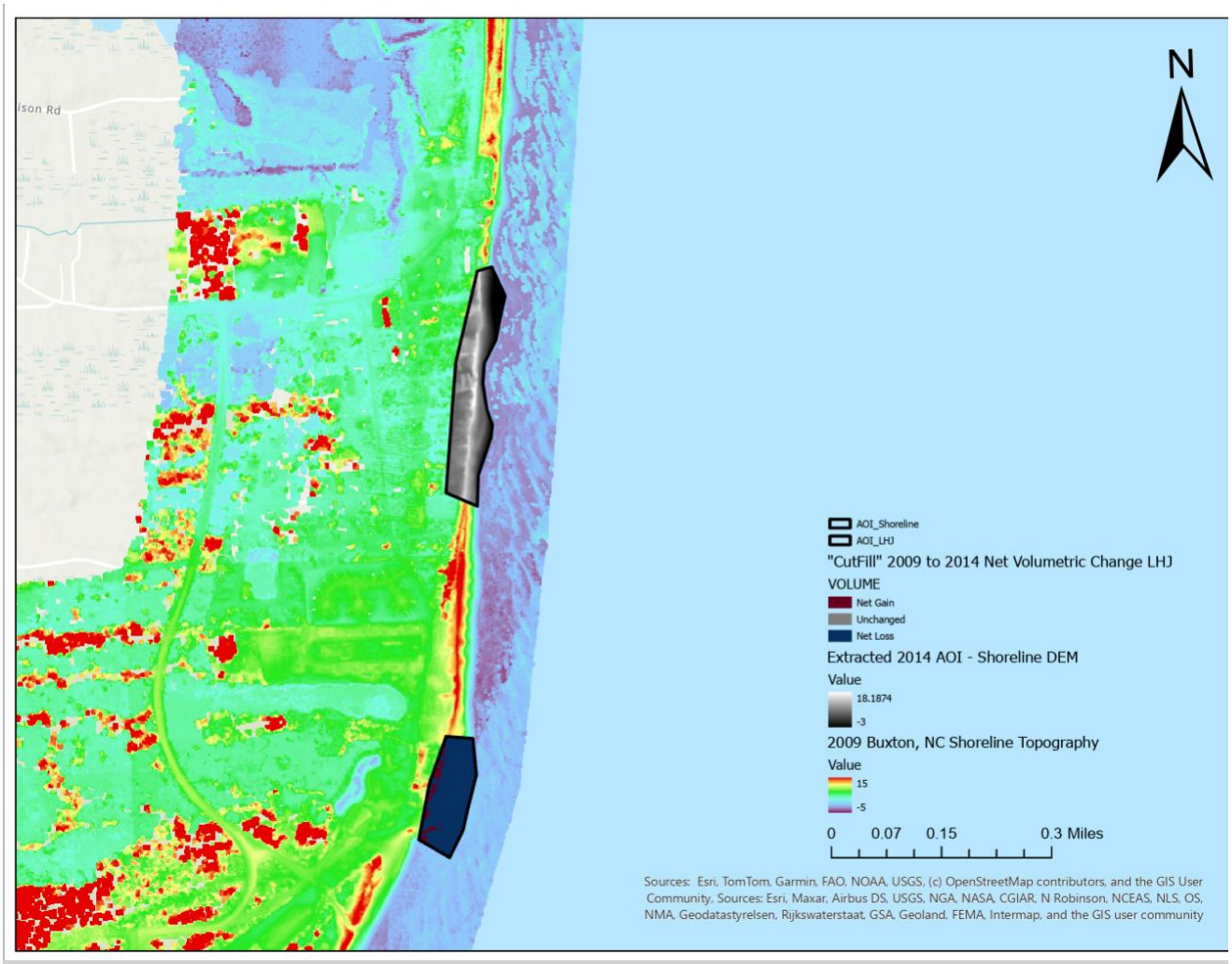

Figure 4: The "Extract by mask" (top) and "CutFill" (bottom) tools respectively. The DEM for a given survey is extracted through the area of interest. Following extraction, CutFill is used to give the difference between two extracted DEM's.

2 – Dimensional Shoreline Analysis

Contours were created at the elevation equals zero level for 2009 and 2019 DEM's. Seven equidistant transects were drawn within both areas of interest latitudinally and the intersection tool was run on all transects to determine where exactly they crossed both elevation equals zero lines. The distance between the intersections for each transect was then measured

and gave a value in feet which is shoreline recession per transect. The transects were then averaged within each zone for cross comparison. Additionally, the fallen houses within the Shoreline AOI were used as a “modern day” elevation equals zero line. The most current attainable DEM for the area was from 2019 and to attempt to account for processes that have occurred on the island over the last 6 years, a polyline was drawn to each point layer on the map which represented a fallen house. This polyline was deemed the current elevation equals zero contour. The same transects were used for this measurement as were used for the 2009 to 2019 shoreline recession transects, and the intersect tool was used to see where the transects crossed the “modern day” elevation equals zero contour and the 2019 elevation equals zero contour. The distance between the intersection of the 2019 contour and 2025 polyline were then measured for each transect and gave a value in feet which is shoreline recession per transect.

Data Analysis

All data extracted from ArcGIS was compiled into an Excel worksheet. Area and volume data were inputted from the Shoreline Zone and LHJ zone for all years. These values were then summed to get net volumetric change for the analysis period. After summing data, the totals were cross compared, analyzed, and visualized for trends. Ultimately it was decided that for this study, two 5-year analysis windows would be investigated (2009-2014 and 2014-2019) along with a decade analysis (2009 to 2019) within the areas of interest to apply a more long-term approach by allowing yearly extremes to balance into the analysis window.

2 – Dimensional shoreline analysis remained a decadal study between 2009 and 2019 but added an extra timeframe of investigation between 2019 and 2024. Data from the shoreline recession methodology was compiled and analyzed in excel.

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Results

Lighthouse Jetty Zone Results:

In between 2009 and 2019, the Lighthouse Jetty Zone has Eroded 865,875 cubic feet of sediment. The area of the LHJ zone is 131,976 feet. On average, the area within the zone lost 6.561 feet of elevation per square foot during the 10-year analysis period. Both five-year terms (2009 – 2014 and 2014 – 2019) had roughly the same amount of net loss, 435,200 cubic feet of loss. (Figure 5)

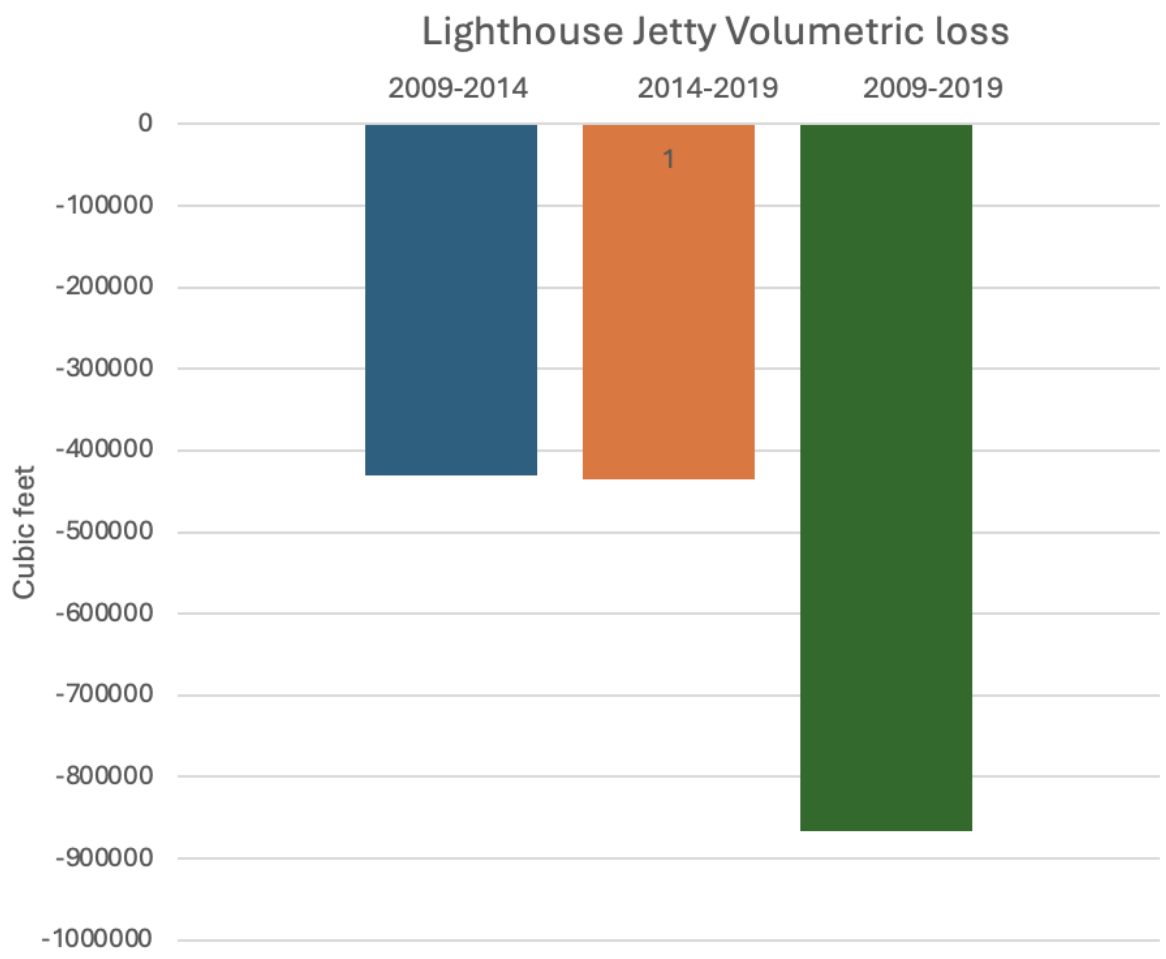

Figure 5: Lighthouse Jetty Zone volumetric change bar graph

Shoreline Zone Results:

In between 2009 and 2019, the Shoreline Zone lost 416,813 cubic feet of sediment. The area of the Shoreline zone is 343,152 square feet. On average, the area within the zone lost 1.215 feet of elevation per square foot during the 10-year analysis. Both five-year terms (2009 – 2014 and 2014 – 2019) had roughly the same amount of net loss, 208,000 cubic feet of loss. (Figure 6)

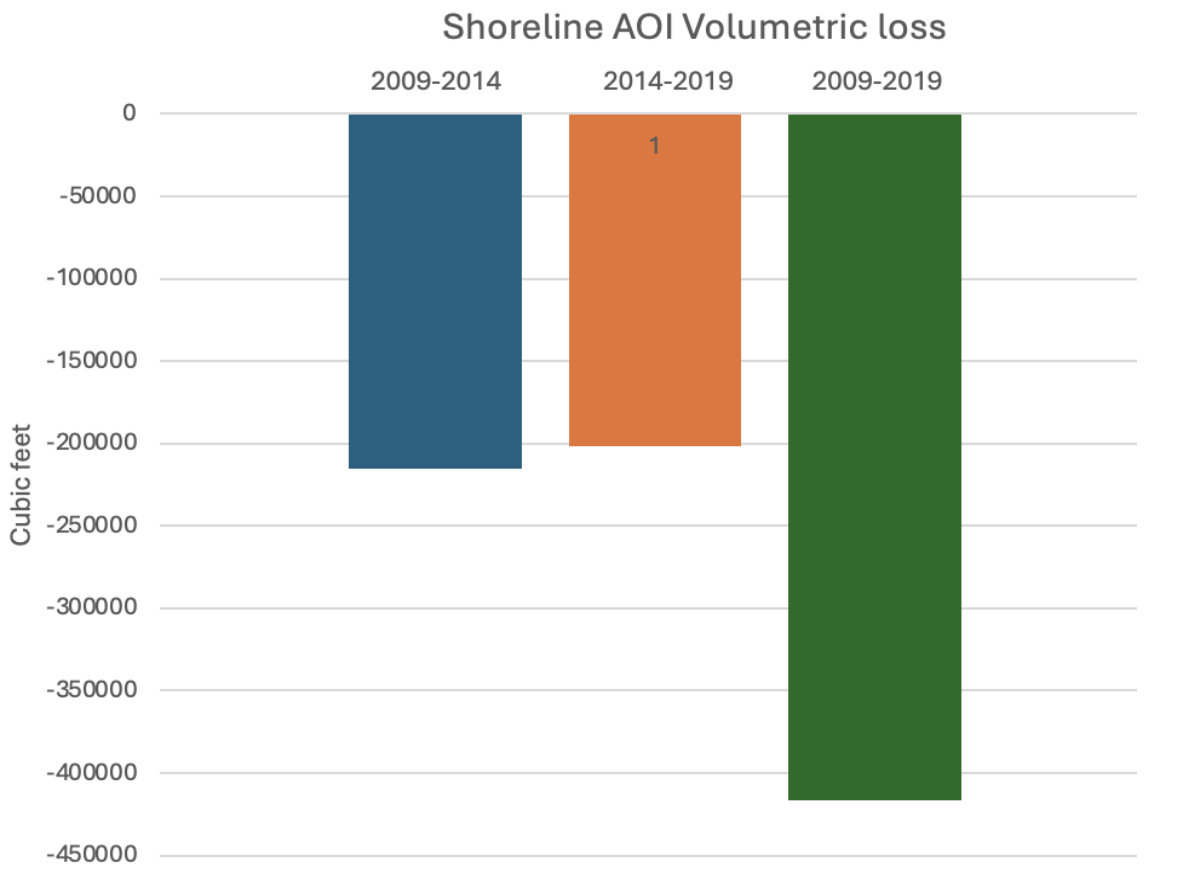

Figure 6: Shoreline Zone volumetric change results bar graph

Shoreline Recession Results:

The Shoreline AOI between 2009 and 2019 saw the lowest linear recession of the elevation equals zero mark at 67 feet (6.7 feet per year). However, in between 2019 and 2025, the Shoreline AOI saw this highest rate of linear recession at 137 feet (22.9 feet per year). The Lighthouse Jetty AOI recessed a total of 136 feet between 2009 and 2019 (13.6 feet per year). (Figure 7)

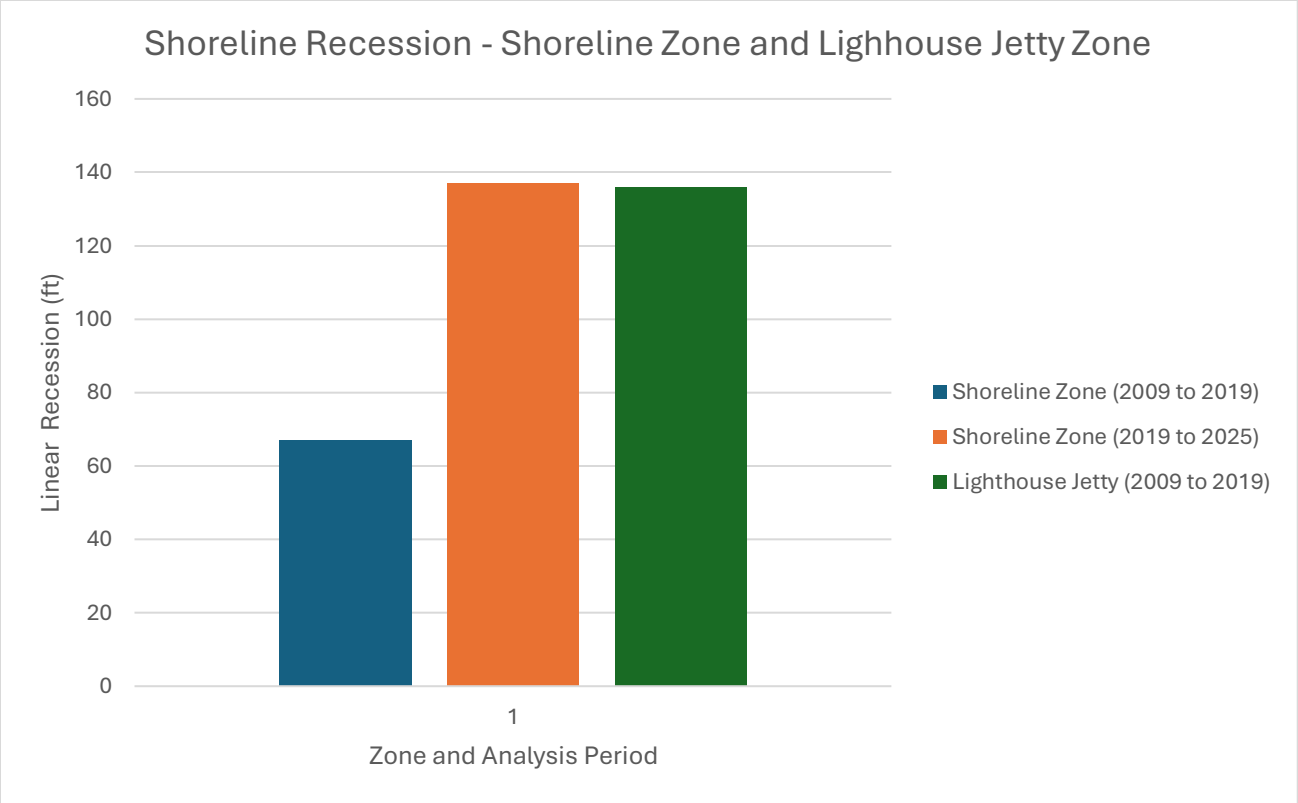

Figure 7: Shoreline recession bar graph for Shoreline Zone 2009 to 2019, Shoreline Zone 2019 to 2025, and LHJ Zone 2009 to 2019, respectively.

Discussion

Volumetric Change

The results from this study demonstrated the consistent erosion along the Buxton, NC shoreline despite community and government efforts such as beach renourishment and creating artificial sand dunes. Despite the Lighthouse Jetty Zone being less than half the area of the Shoreline Zone, it eroded nearly double that of the Shoreline AOI during the decade of analysis.

The natural landscape of Lighthouse Beach consisted of a once tall dune line followed by dense maritime forest vegetation. Extruding from the dune line were three groins which were built to protect the lighthouse and a nearby naval base from erosion. However, with the erosion that occurred post-groin failure, the tall dune line and eastward extruding beach which expanded to the north of the groin has eroded.

In the early 2000's the Buxton community began to feel more effects of erosion occurring in the area. Shoreface erosion caused the NPS to move Cape Hatteras Lighthouse hundreds of yards inland from its original position. No work was able to be done to the failing groins due to the ban on hard structures. Additionally, the lighthouse movement was funded by the government and NPS, locals and homeowners in the area were not so fortunate. The cost of relocation by itself is high, not including any extra plots of land the homeowner may have to purchase to successfully relocate or the possibility of the home being physically impossible to move due to structural restrictions or neighboring homes blocking the large amount of space needed for a project as complex as moving a house on a sandy substrate. Correlatively, after the erosion of Lighthouse Beach, areas to the north began to see greater rates of shoreline recession and a higher number of house falls. This correlation points to the idea that mass volumetric and elevation loss within the Lighthouse Jetty Zone was the loss of the expanded and stabilized beach strand created by the 3 groins that were placed there in the mid 20th century and creates a space for future research to investigate mechanisms for this erosion.

Interpretive analysis – Proposed Erosional Mechanism

After examining the data and figures generated from this study, a possible mechanism needs to be investigated in future research. One, which is supported by the results in this study, is that a residual sediment buffer remained during the groins deterioration. After the groins completely failed at Lighthouse Beach, the sediment buffer acted as a temporary groin by extruding eastward into the Atlantic Ocean with a relatively high elevation when compared to other areas along the local shoreline. This extrusion of the beach strand was capable of temporarily holding upstream sediment and protecting houses north of the groins even during the late stages of structural failure. However, once this area completely eroded to parallel with the northern stretch of beach strand, the southerly flow of sediment was uninhibited and resulted in mass shoreline recession within the shoreline zone. In turn, causing homes to become susceptible to the surf zone and ultimately fall. The groin deterioration and large-scale volumetric loss at Lighthouse beach was a precursor and predictor for the significantly higher rates of shoreline recession within the Shoreline Zone between 2019 and 2025.

2-Dimensional Shoreline Recession

The Lighthouse Jetty Zone over doubled the recession of the shoreline zone during the 2009 to 2019 analysis window. With this came major volumetric loss. Within the initial decade long period of investigation, the Shoreline Zone experienced much lower rates of shoreline recession (6.7 feet per year) when compared to the LHJ Zone (13.6 feet per year). However, when the period of analysis was set to 2019 – 2025, shoreline recession in the Shoreline AOI almost tripled to 22.9 feet per year. As one may expect, the higher rates of shoreline recession in the residential area correlates with the time when house falls began to increase. When visually observing the DEM's that were created at the start of the project, it is reasonable to suggest that

the shoreline recession seen in the more northern AOI increased as the LHJ Zone decreased in volume and recessed linearly to a state in which there is no more shoreface extrusion at Lighthouse Beach.

Conclusion

While the conclusions of this study are limited, the stage is set for future research to pinpoint a mechanism for the increased rates of erosion and house falls along the eastern Buxton shoreface. The mechanism which seems most reasonable within the results and discussions of this research is that the deteriorated groins on Lighthouse Beach led to the destabilization of the immediately surrounding beach strand. Lighthouse beach ultimately recessed both volumetrically and linearly to a point that it could no longer act as an artificial stabilizer soon after the groin failed, which ultimately led to the destabilization of upstream areas.

If this were to be the case, local and state governments along with the NPS need to immediately open the case for structural improvements for specifically the first groin at Lighthouse Beach. If a direct relationship between the loss of the groin and its residual sediment and their impacts on the residential areas to the north of Lighthouse beach, it would create a case to refurbish the groin and attempt to restabilize the shoreline of Buxton, NC.

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